MODULE - 04

PUSH DOWN AUTOMATA (PDA)

- PDA can be viewed as the finite automata with stack.
- An array stack provides memory & increases the capacity of the m/c.
- PDA can do the following
  (1) Read I/P symbol
  (2) PUSH operation
  (3) Check empty condition of a stack through an initial stack symbol
  (4) Read top symbol of a stack
  (5) Make state changes
- PDA is more powerful than FA.
- PDA is as shown in Fig. below

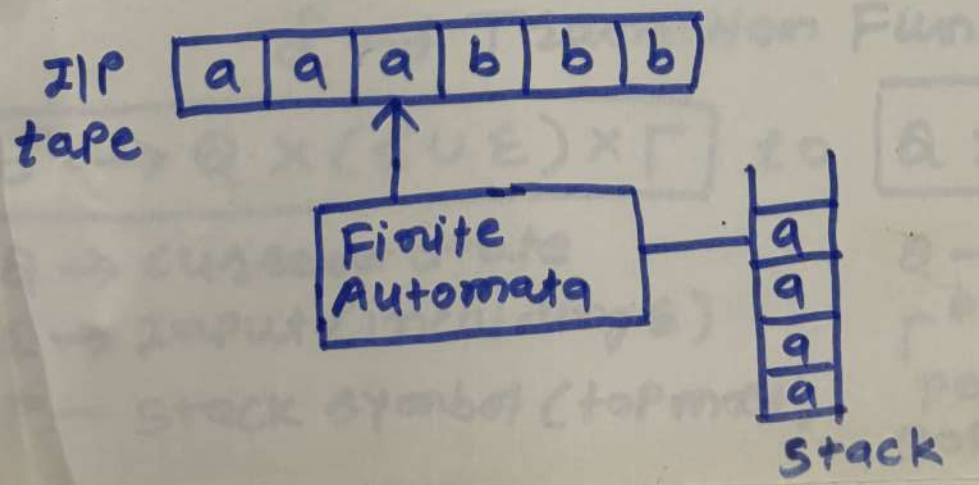


Fig. PDA

DA Uses 3 stack operations

1) PUSH - It inserts a symbol onto the top of the stack.
2) POP - It removes the top symbol from the stack.
(3) NOP - It nothing on stack

Representation of PDA -

- PDA is represented by 7-tuples

$$M = (Q, \Sigma, \delta, q_0, F, \Gamma, Z_0)$$

Where $q_0 \rightarrow$ Initial state symbol
$\Sigma \rightarrow$ Finite I/P symbols
$Q \rightarrow$ Finite set of states
$F \rightarrow$ Final state
$\Gamma \rightarrow$ Stack symbol
$Z_0 \rightarrow$ An initial stack symbol
$\delta \rightarrow$ Transition Function

$\delta \rightarrow Q \times (\Sigma \cup \varepsilon) \times \Gamma$ to $Q \times \Gamma^*$

$Q \rightarrow$ current state
$\Sigma \rightarrow$ Input (including ε)
Γ - Stack symbol (topmost)

$Q \rightarrow$ Next state
$\Gamma^* \rightarrow$ It can perform PUSH, POP, NOP on stack

Transition Behaviour -

PUSH

$\delta(q_1, a, b) = (q_2, ab)$

- q_1 → Current State
- a → I/P symbol
- b → Stack symbol
- q_2 → Next State
- ab → Stack symbol b is replaced with ab



② POP

$\delta(q_1, a, b) = (q_2, \varepsilon)$

- q_1 → Current State
- a → I/P symbol
- b → Stack symbol
- q_2 → Next State
- ε → Stack symbol b is replaced with ε



③ NOP

$\delta(q_1, a, b) = (q_2, b)$

- q_1 → Current State
- a → I/P symbol
- b → Stack symbol
- q_2 → Next State
- b → Stack symbol b is replaced with b



PUSH DOWN AUTOMATA

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~~① Construct PDA for the language $L = \{a^n b^{2n} \mid n \geq 1\}$~~

① Design PDA to recognize $L = \{a^n b^n / n \geq 1\}$

⇒ Step 1 - [Theory] . To design PDA to recognize the language $\{a^n b^n \mid n \geq 1\}$

Step : [Logic]

For each 'a' PUSH 1 X
For each 'b' POP 1 X

Step 3 : [Implementation]

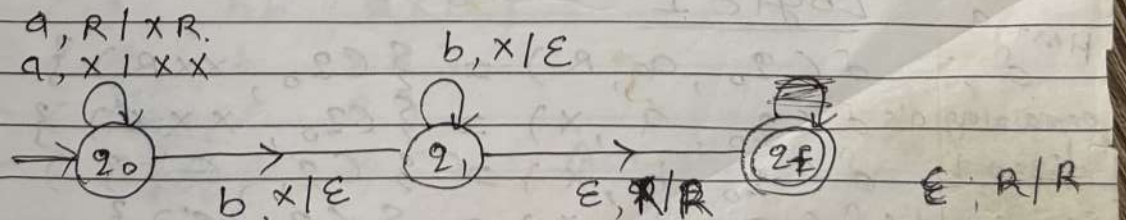
$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$

Where

$Q = \{q_0, q_1, q_f\}$
$\Sigma = \{a, b\}$
$\Gamma = \{X, R\}$
$F = \{q_f\}$
$q_0 = q_0$
$z_0 = R$

δ :-

first a $\delta(q_0, a, R) = \{(q_0, XR)\}$
rem a's $\delta(q_0, a, X) = \{(q_0, XX)\}$
first b $\delta(q_0, b, X) = \{(q_1, \epsilon)\}$
rem b's $\delta(q_1, b, X) = \{(q_1, \epsilon)\}$
final $\delta(q_1, \epsilon, R) = \{(q_f, R)\}$



a, R/XR ⇒ a → Input symbol
R → Top of the stack
XR → New stack contents after PUSH

Step 4 : Instantaneous Descriptions

Exam

$(q_0, aaaabbbb, R)$
$\vdash (q_0, aaabbbb, XR)$
$\vdash (q_0, aabbbb, XXR)$
$\vdash (q_0, abbbb, XXXR)$
$\vdash (q_0, bbbb, XXXXR)$
$\vdash (q_1, bbb, XXXR)$
$\vdash (q_1, bb, XXR)$
$\vdash (q_1, b, XR)$
$\vdash (q_1, \in, R)$
$\vdash (q_f, \in, R)$ Accept

$(q_0, abab, R)$
$\vdash (q_0, bab, XR)$
$\vdash (q_1, ab, R)$ Reject

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(5) Design PDA to recognize
$L = \{ a^n b^{n+1} \mid n \geq 1 \}$

aaaa b bbbb

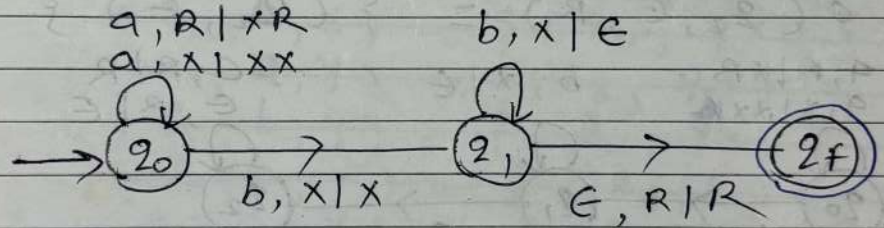
⇒ Step 2 - Logic

For each 'a' Push 1X
For first b perform no operation just change the state
For each 'b' Pop 1X

Step 3 - δ :

First a $\delta(q_0, a, R) = \{(q_0, XR)\}$
rem a's $\delta(q_0, a, X) = \{(q_0, XX)\}$
first b $\delta(q_0, b, X) = \{(q_1, \in)\}$
rem b's $\delta(q_1, b, X) = \{(q_1, \in)\}$

final $\delta(q_1, \in, R) = \{(q_f, R)\}$



Step - 4 -

$(q_0, aaaabbbbbb, R)$
$\vdash (q_0, aaabbbbbb, XR)$
$\vdash (q_0, aabbbbbb, XXR)$
$\vdash (q_0, abbbbbb, XXXR)$
$\vdash (q_0, bbbbbb, XXXXR)$
$\vdash (q_1, bbbbb, XXXXR)$
$\vdash (q_1, bbb, XXXR)$
$\vdash (q_1, bb, XXR)$
$\vdash (q_1, b, XR)$
$\vdash (q_1, \in, R)$
$\vdash (q_f, \in, R)$

PDA by Null Store Method

(6) Design PDA to recognize $L = \{a^m b^m c^n \mid m, n \geq 1\}$ by null store method

$\Rightarrow$ $m = 2, n = 2$

aabbcc

Step 2 - Logic

For each a push 1X

For each b pop 1X

Bypass all the c's

Step 3 - δ :

first a $\delta(q_0, a, R) = \{(q_0, XR)\}$

rem a's $\delta(q_0, a, X) = \{(q_0, XX)\}$

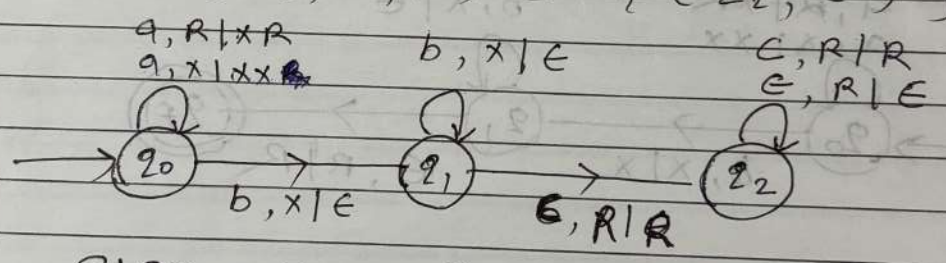
first b $\delta(q_0, b, X) = \{(q_1, \epsilon)\}$

rem b's $\delta(q_1, b, X) = \{(q_1, \epsilon)\}$

first ϵ $\delta(q_1, c, R) = \{(q_2, R)\}$

rem c's $\delta(q_2, c, R) = \{(q_2, R)\}$

$\delta(q_2, \epsilon, R) = \{(q_2, \epsilon)\}$



Step - 4 - Example

$(q_0, aabbcc, R)$

$\vdash (q_0, abbcc, XR)$

$\vdash (q_0, bbcc, XXR)$

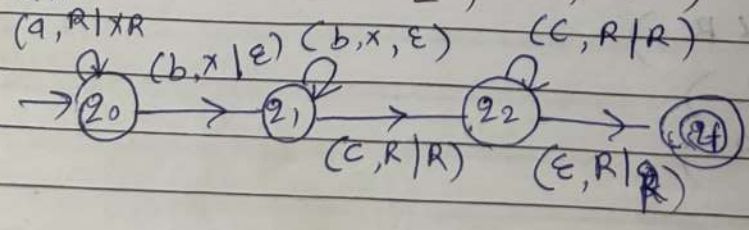
$\vdash (q_1, bcc, XR)$

$\vdash (q_1, cc, R)$

$\vdash (q_2, c, R)$

$\vdash (q_2, \epsilon, R)$

$\vdash (q_2, \epsilon, \epsilon)$

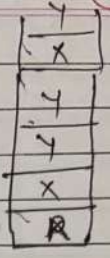


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(7) Design PDA to recognize
$L = \{ wcw^R \mid w \in \{a,b\}^* \}$



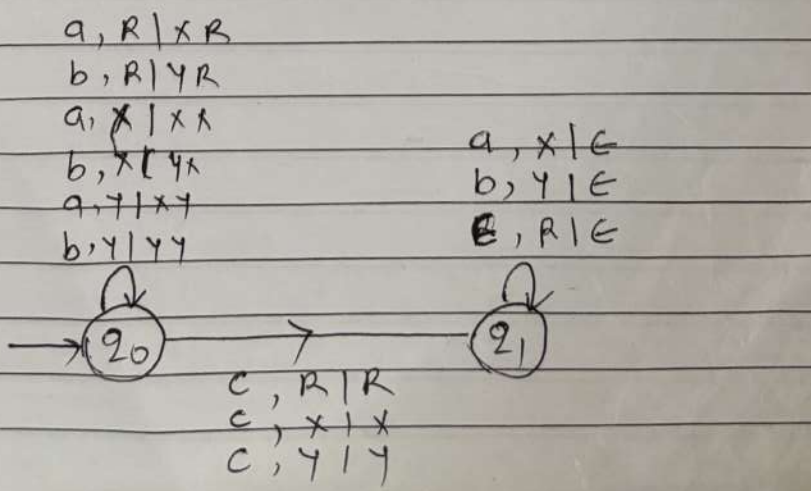
⇒ w^R = Reverse of w
abbabCbabba

Step 2 - Logic
For each a push 1X
for each b push 1Y
for C perform no operation, just change the state
Now for each a pop 1X
Now for each b pop 1Y

Step 3 - δ :

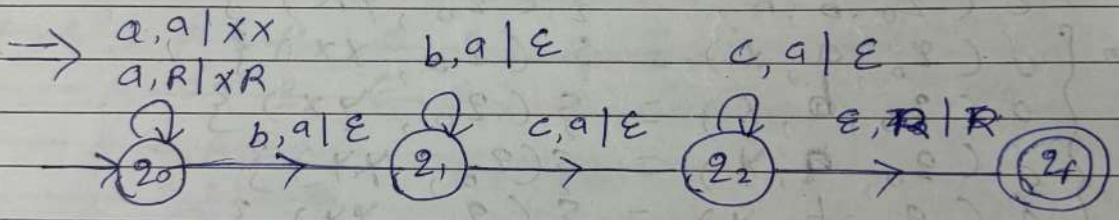
$\delta(q_0, a, R) = \{(q_0, XR)\}$
$\delta(q_0, b, R) = \{(q_0, YR)\}$
$\delta(q_0, a, X) = \{(q_0, XX)\}$
$\delta(q_0, b, X) = \{(q_0, YX)\}$
$\delta(q_0, a, Y) = \{(q_0, XY)\}$
$\delta(q_0, b, Y) = \{(q_0, YY)\}$
$\delta(q_0, c, R) = \{(q_1, R)\}$
$\delta(q_0, c, X) = \{(q_1, X)\}$
$\delta(q_0, c, Y) = \{(q_1, Y)\}$

$\delta(q_1, a, X) = \{(q_1, \epsilon)\}$
$\delta(q_1, b, Y) = \{(q_1, \epsilon)\}$
$\delta(q_1, \epsilon, R) = \{(q_1, \epsilon)\}$



$(q_0, abbabcbabba, R)$
$\vdash (q_0, bbabcbabba, XR)$
$\vdash (q_0, babcbabba, YXR)$
$\vdash (q_0, abcbabba, YYXR)$
$\vdash (q_0, bcbabba, XYYXR)$
$\vdash (q_0, cbabba, YXYYXR)$
$\vdash (q_1, babba, YXYYXR)$
$\vdash (q_1, abba, XYYXR)$
$\vdash (q_1, bba, YYXR)$
$\vdash (q_1, ba, YXR)$
$\vdash (q_1, a, XR)$
$\vdash (q_1, \epsilon, R)$
$\vdash (q_1, \epsilon, \epsilon)$

(8) $a^{m+n} b^m c^n \mid m, n \geq 1$



* $L = \{ a^{2n} b^n \mid n \geq 1 \}$ aaaabb

Logic →
for first a ~~skip~~ push 1X
for second a ~~push~~ ~~1X~~ skip
third a ~~skip~~ push 1X
fourth a ~~push~~ skip
for each b pop 1 X

$\delta(q_0, a, R) = \delta(q_0 X R)$
$\delta(q_0, a, R) = \delta(q_1, XR)$

$\delta(q_1, a, X) = \delta(q_0, XX)$
$\delta(q_0, a, X) = \delta(q_1, XXX)$

first b $\delta(q_1, b, X) = \delta(q_2, \varepsilon)$
~~$\delta(q_1, X)$~~
remaining b $\delta(q_2, b, X) = \delta(q_2, \varepsilon)$

$\delta(q_2, \varepsilon, R) = \delta(q_f, R)$

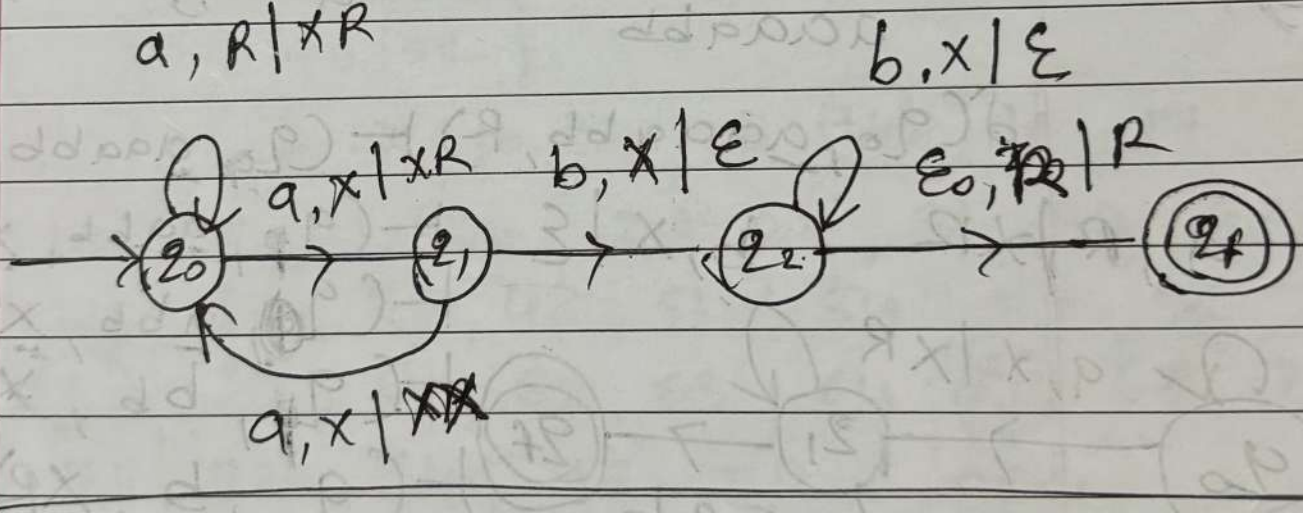
$\delta(q_0, aabb, R)$

$\delta(q_0, abb, XR)$

$\delta(q_1, bb, XR)$

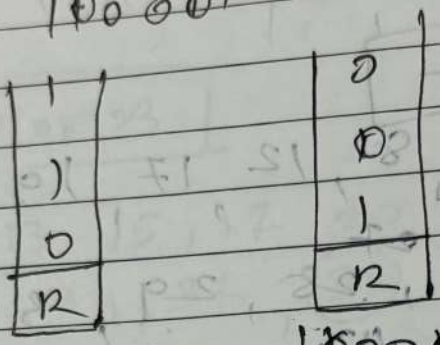
$\delta(q_1, b, R)$

$\delta($



* Construct NDPDA for
$L = \{ wwR \mid w \text{ is } (0+1)^+ \}$
(even Palendrome)

⇒ 01110
10001



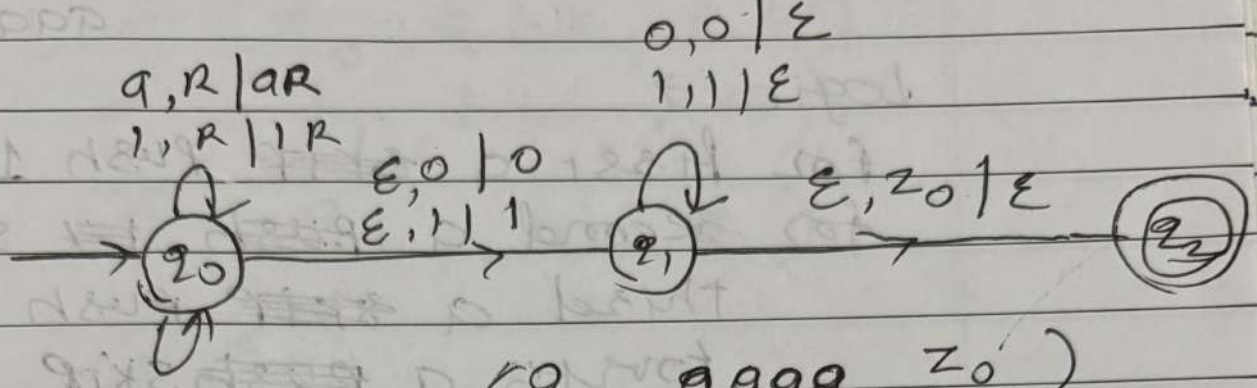
01110 10001

1) $\delta(q_0, 0, R) = (q_0, 0R)$
$\delta(q_0, 1, R) = (q_0, 1R)$
$\delta(q_0, 1, 0) = (q_0, 10)$
$\delta(q_0, 0, 0) = (q_0, 00)$
$\delta(q_0, 0, 1) = (q_0, 01)$
$\delta(q_0, 1, 1) = (q_0, 11)$

2) $\delta(q_0, \varepsilon, 0) = (q_1, 0)$
$\delta(q_0, \varepsilon, 1) = (q_1, 1)$

3) $\delta(q_1, 0, 0) = (q_1, \varepsilon)$
$\delta(q_1, 1, 1) = (q_1, \varepsilon)$

4) $\delta(q_1, \varepsilon, R) = \delta(q_f, R)$



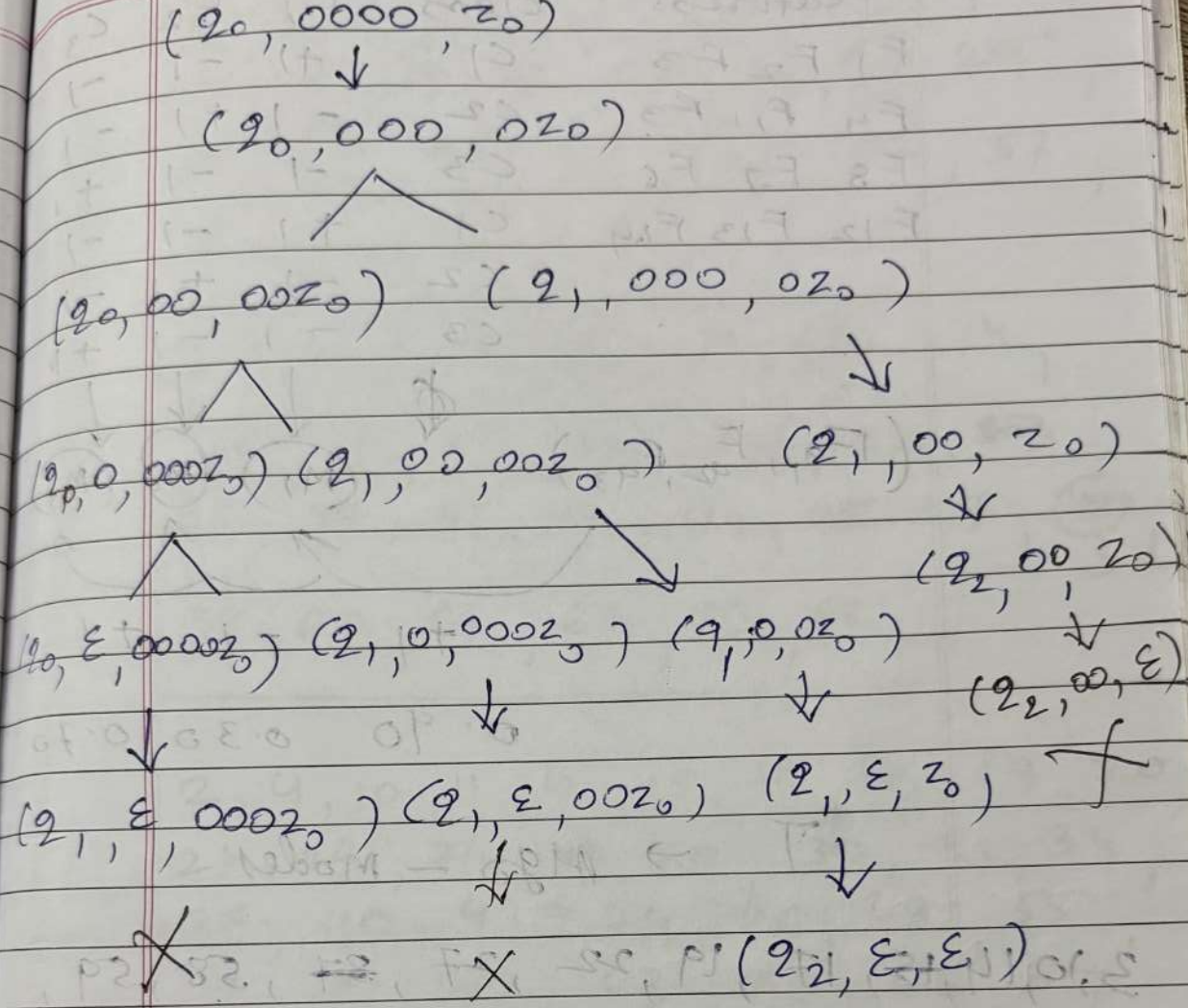
1,0 | 10
0,0 | 00
0,1 / 0,1
1,1 | 11

$(q_0, aaaa, Z_0)$
$\vdash (q_0, 000, 0Z_0)$
$\vdash$ NC ~~$(q_0, 00, 00)$~~ C

$(q_0, 00, 00)$ $(q_1, 00, Z_0)$

NC ⊢ C

$(q_0, 0, 000Z_0)$ 4



$(q_0, 0000, z_0) \vdash (q_0, 000, 0z_0)$
$\vdash (q_0, 00, 00z_0)$
$\vdash (q_1, 00, 00z_0)$
$\vdash (q_1, 0, 0z_0)$
$\vdash (q_1, \varepsilon, z_0)$
$\vdash (q_2, \varepsilon, \varepsilon)$

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Theory

What is NDPDA, Difference between FA & PDA.
Difference between PDA & NPDA
Representation of PDA
What is PDA. Transition function of PDA.
Different operations on PDA.
Machine